

CSCI 460—Operating Systems

Lecture 01

Textbook: Operating Systems — Internals and Design Principles (9th edition) by William Stallings

- Course Homepage:
 - <https://montana.instructure.com/courses/35966>
- Course schedule:
 - <https://github.com/adiesha/CSCI460Fall2026>
- Basic operating systems (2/3) | Advanced operating systems (1/3)
- Basic operating systems
 - Memory management:
 - Paging
 - Memory hierarchy
 - Processor management:
 - Basic concepts: Processor, process, job, etc.
 - Process scheduler.
 - Scheduling policy.
 - Deadlock concepts, handling deadlocks, avoidance, prevention, etc.
 - Starvation.
 - Device management:
 - Device handler strategies, etc.
 - File management:
 - Basic definitions.
 - File organization.
 - Access control.
- Advanced operating systems
 - Threads
 - How to create threads in Linux.
 - Symmetric multiprocessing
 - Multiprocessor scheduling
 - Mutual exclusion.
 - Algorithms for mutual exclusion (centralized, distributed etc.).
 - Real-time scheduling.
 - Networking

- Distributed Mutual Exclusion Concepts.
- Security
 - Diffie-Hellman algorithm
 - RSA
- Focus more on concepts and algorithms of OS and how OS works.
- We will use c programs under Linux operating system to understand concepts.
- You are required to read the sections of the book that is assigned.
- Canvas is the main repository of lecture notes, assignments, and solutions, but I will keep the schedule in GitHub page.
- Canvas will be used for announcements. Please check the Canvas for announcements.
- Evaluation
 - Random Pop-up tests (in class) 5% (I will allow 1 to be dropped)
 - 5 homework (CANVAS quizzes) 20%
 - 3 in class tests (30%)
 - 3 programming assignments (24%)
 - 1 final project (16%)
 - Attendance (5%)

1. Name some operating systems (OS) that you have heard of

a. _____

b. _____

c. _____

d. _____

e. _____

f. _____

g. _____

h. _____

- i. _____
- j. _____

Computer system overview

- You already have an idea about the overview of the computer system from the computer systems class (CSCI 366).
- Here I will touch briefly about the computer system overview.

Basic Elements

- Processor: Controls the operation of the computer and performs its data processing functions.
- Main memory: Controls the operation of the computer and performs its data processing functions.
- I/O modules: Move data between the computer and its external environment.
- System Bus: Provides communication among processors, main memory, and I/O modules.

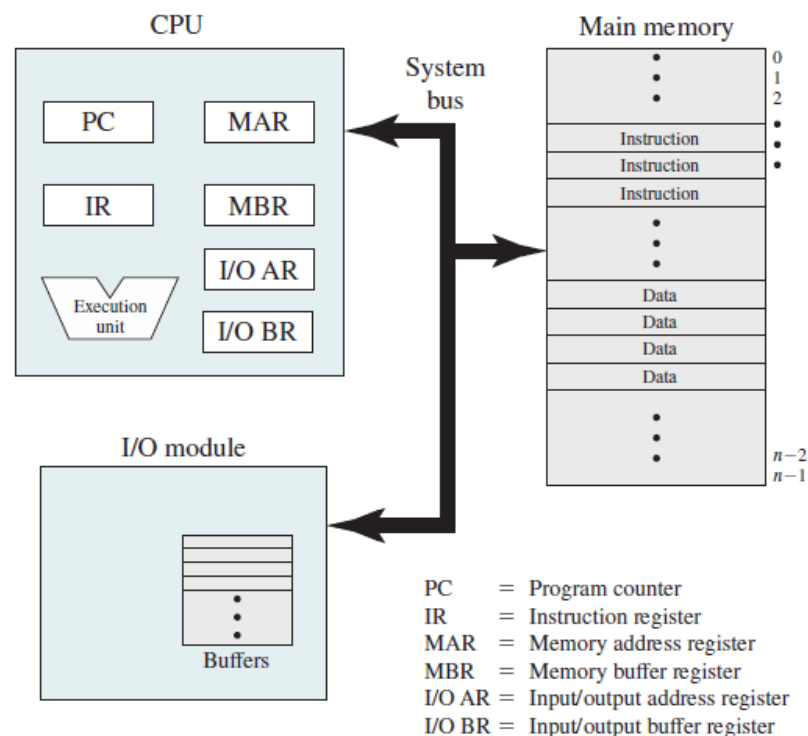


Figure 1.1 Computer Components: Top-Level View

Instruction cycle

- Instruction processing consists of two steps:
 - Fetches the instruction from the memory one at a time.
 - Executes the instruction.
- Program execution consists of repeating the process of instruction fetch and instruction execution.
- Processing required for single instruction is called an instruction cycle.
- Program execution halts only if the processor is turned off, some sort of unrecoverable error occurs, or a program instruction that halts the processor is encountered.

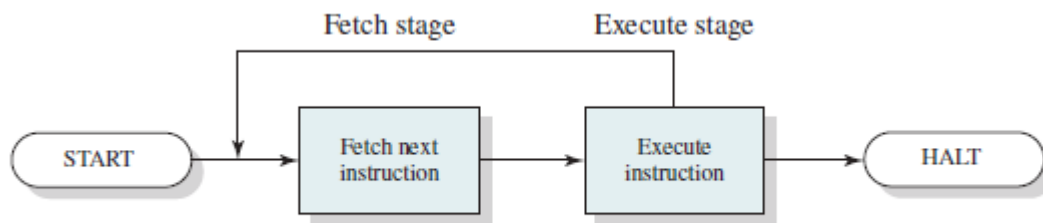


Figure 1.2 Basic Instruction Cycle

- Typically, the program counter (PC) holds the address of the next instruction to be fetched.
- Processor always increments the PC after each instruction fetch cycle unless instructed otherwise.
- The fetched instruction is loaded into the instruction register (IR)
 - The instruction contains bits that specify the action the processor is to take.
- Typically, these actions fall into 4 categories:
 - Processor $\leftrightarrow$ memory: Data transfer between processor and memory (both ways)
 - Processor $\leftrightarrow$ I/O: Data transfer between to or from a peripheral device by transferring between processor and I/O module.
 - Data Processing: Arithmetic and logical operations on Data.
 - Control: An instruction may specify that sequence of execution be altered.

Interrupts

- Virtually all computers provide a mechanism by which other modules (I/O, memory) may interrupt the normal sequencing of the processor.
- There are several types of Interrupts:

Table 1.1 Classes of Interrupts

Program	Generated by some condition that occurs as a result of an instruction execution, such as arithmetic overflow, division by zero, attempt to execute an illegal machine instruction, or reference outside a user's allowed memory space.
Timer	Generated by a timer within the processor. This allows the operating system to perform certain functions on a regular basis.
I/O	Generated by an I/O controller, to signal normal completion of an operation or to signal a variety of error conditions.
Hardware failure	Generated by a failure, such as power failure or memory parity error.

- Improves processor utilization.
 - I/O devices are much slower than the processor.
- When an interrupt occurs user program is suspended, and the interrupt handler code is executed, once the interrupt handler code is finished then the execution of the suspended process resumes.
- The **user program does not have to contain any special code to accommodate interrupts; the processor and the OS are responsible** for suspending the user program, then resuming it at the same point.
- The interrupt-handler routine is generally part of the OS.
- To accommodate interrupts, an interrupt stage is added to the instruction cycle.

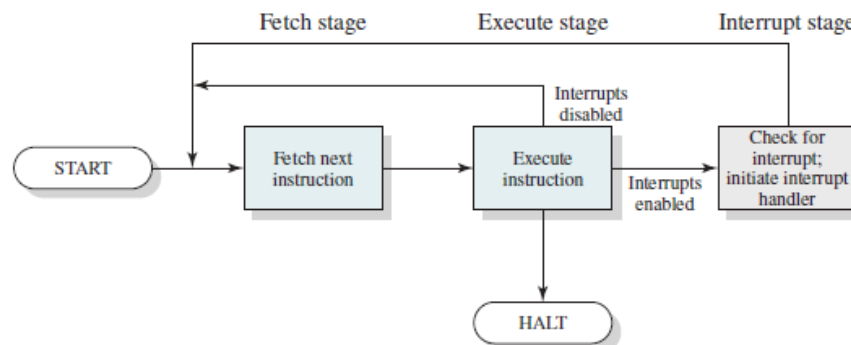


Figure 1.7 Instruction Cycle with Interrupts

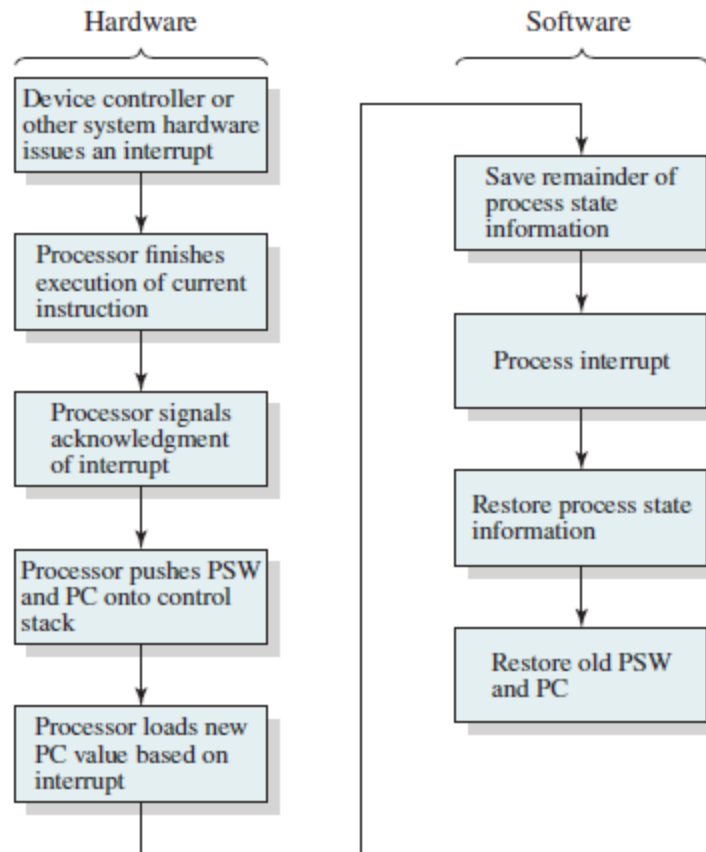


Figure 1.10 Simple Interrupt Processing

- The PSW contains status information about the currently running process, including memory usage information, condition codes, and other status information such as an interrupt enable/disable bit and a kernel/user-mode bit.

What is an OS?

- OS is the computing system that manages all the hardware and software.
- In the first part of this course, we will mainly focus on how OS works, the related concepts as well as algorithms.
- Some concepts related to distributed computing and network operating systems will be covered later in the course.
- Some concepts on computer architecture and hardware will be covered as well.

What is OS composed of?

1. Memory Manager
 - a. Controls and manages the main memory.
2. Processor Manager
 - a. Decides how to allocate processing power to processes.
3. Device Manager
 - a. Manages every device, channel, and control unit.
4. File Manager
 - a. Manages file system.
5. Network Manager
 - a. Manages network connections.